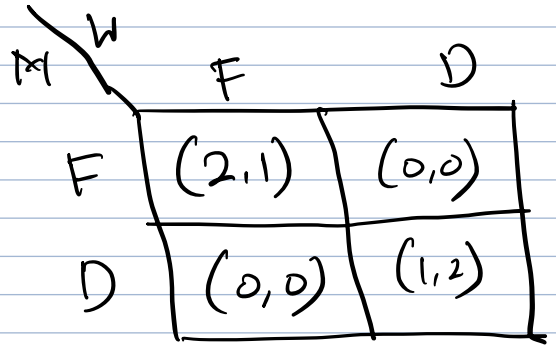
Class 7

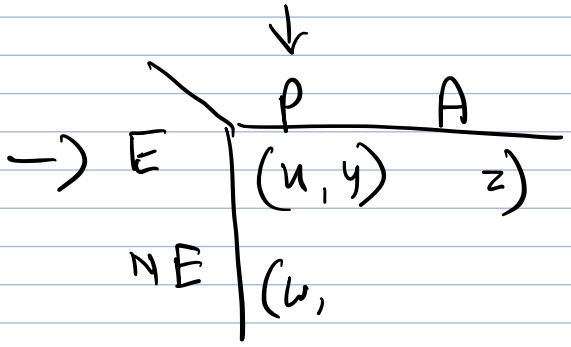
Nash Equilibrium

Defn For a game $G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$, a strategy profile s^* is a Nash Equilibrium if $\forall i \in N$ and $\forall s_i \in S_i$

$$u_i(s^*) \geq u_i(s_i, s_{-i}^*)$$



(F, F) and (D, D) are the NE



$y \geq z$

$u \geq w$

Best reply Correspondance $b_i : S_{-i} \rightrightarrows S_i$

function $\bar{b}_i : S_{-i} \to 2^{S_i}$

$$b_i(s_{-i}) = \arg\max_{t_i \in S_i} u_i(t_i, s_{-i})$$

Nash Equilibrium A profile s^* is an NE if

$\forall i \in N$, $s_i^* \in b_i(s_{-i}^*)$.

Beauty contest game

$(0,0,0)$ is an NE?

- $N = \{1,2,3\}$
- $S_i = [0, 100]$

$$T(x_1, x_2, x_3) = \frac{2}{3}\left(\frac{1}{3}\sum_{i=1}^{3} x_i\right)$$

$$W(x_1, x_2, x_3) = \arg\min_{i \in N} \left| x_i - T(x_1, x_2, x_3) \right|$$

$$u_i(x_1, x_2, x_3) = \begin{cases} 0 & \text{if } i \notin W(x_1, x_2, x_3) \\ \frac{1}{|W(x_1, x_2, x_3)|} & i \in W(x_1, x_2, x_3) \end{cases}$$

1) (a, a, a) where $a \in [0, 100]$ is an NE ✗

Find all the NE of this game?

Second price Auction game

$(v_1, v_2, \ldots, v_n)$ is an NE

$(v_1, 0, 0, \ldots, 0)$ is an NE $\quad u_i(b_1, \ldots, b_n)$

$v_1 = 10$

$v_2 = 11$

$$= \begin{cases} 0 & b_i \neq \max_{j \in N} b_j \\ v_i - \max_{j \neq i} b_j & b_i = \max_{j \in N} b_j \\ \left| \{k \mid b_k = \max_j b_j\} \right| & \end{cases}$$

Find all NE of the game.

Example

a_1 a_2

$$p(a_1, a_2) = 2 - a_1 - a_2$$

c_1 , c_2

- $N = \{F_1, F_2\}$
- $S_i = [0, 1] \quad \forall i$
- $u_i(a_1, a_2) = (2 - a_1 - a_2) a_i - c_i a_i$

(a_1^*, a_2^*)

$$u_1(a_1^*, a_2^*) \geq u_1(a_1, a_2^*)$$

$$u_2(a_1^*, a_2^*) \geq u_2(a_1^*, a_2)$$

$$b_1(\bar{a}_2) = \arg\max_{a_1 \in [0,1]} (2 - a_1 - \bar{a}_2) a_1 - c_1 a_1$$

$$= \frac{2 - c_1 - \bar{a}_2}{2}$$

$$b_2(\bar{a}_1) = \frac{2 - c_2 - \bar{a}_1}{2}$$

$$(2 - a_1 - \bar{a}_2) + a_1(-1) - c_1 = 0$$

$$a_1 = \frac{2 - c_1 - \bar{a}_2}{2}$$

(a_1^*, a_2^*) is a NE

$a_1^* \in b_1(a_2^*) \qquad a_2^* \in b_2(a_1^*)$

$$a_1^* = \frac{2 - c_1 - a_2^*}{2} \qquad a_2^* = \frac{2 - c_2 - a_1^*}{2}$$

$$a_1^* = \frac{2 - 2c_1 + c_2}{3} \qquad a_2^* = \frac{2 - 2c_2 + c_1}{3}$$

$$u_1(a_1^*, a_2^*) = (a_1^*)^2$$

$$u_2(a_1^*, a_2^*) = (a_2^*)^2$$